

MAT 380: Exam 1

March 6, 2026

Name: _____

I wish you the best of luck! Some things to keep in mind with the exam:

- Please use complete sentences and proper formatting/notation when writing proofs. Reference any named theorem or axiom you use by name.
- You are encouraged to use scratch paper for scratch work before writing a proof if you wish, but I will not grade your scratch work.
- You can get partial credit for writing definitions/what you know about a problem if you don't know how to solve it.
- Pay attention to your points total and progress. Answer questions you know how to solve first, and move on if you are struggling. No question is worth more than 10 points.

Have a good weekend!

The exam will take **1 hour** and is **60 total points**.

1. (5 points each) Give an example of each of the following, or briefly argue why such a request is impossible.

- (a) An uncountable collection of disjoint open intervals.

Not possible. Because $\mathbb{Q}$ is dense in $\mathbb{R}$, each open interval (a, b) contains a unique element $q \in \mathbb{Q}$ such that $a < q < b$. But $\mathbb{Q}$ is countable, so there can only be countably many disjoint open intervals.

- (b) A sequence with an infinite number of ones that converges to a limit not equal to one.

Not possible. If it converged to a limit $L \neq 1$, then if $\varepsilon < |1 - L|$, it wouldn't be possible to find an N such that all terms a_n in the sequence with $n \geq N$ were within ε of L , since infinitely many terms are equal to 1.

- (c) A sequence that contains subsequences converging to every point in the infinite set

$$\{1, 1/2, 1/3, 1/4, 1/5, \dots\}$$

$$(a_n) = 1, \frac{1}{2}, 1, \frac{1}{2}, \frac{1}{3}, 1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, 1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \frac{1}{5}, 1, \dots$$

- (d) A sequence with infinitely many peak terms that is not monotone.

Definition: Given a sequence (x_n) , a *peak term* is a term x_m such that $x_m \geq x_n$ for all $n \geq m$.

$$(a_n) = 0, 1, 0, \frac{1}{2}, 0, \frac{1}{3}, 0, \frac{1}{4}, 0, \frac{1}{5}, 0, \dots$$

2. (5 points) Why do we not need to include the existence of greatest lower bounds in the Axiom of Completeness? I.e, explain why the existence of a supremum for any set that is bounded above implies the existence of an infimum for any set that is bounded below. (No full proof needed, just an explanation.)

Suppose we have a set A that is bounded below, and let B be the set of all lower bounds for A . Then the supremum of set B is the greatest lower bound/infimum for set A .

3. (10 points) Prove that the limit superior exists for any bounded sequence. I.e. prove that for any bounded sequence (a_n) , the sequence (y_n) defined by $y_n = \sup\{a_k : k \geq n\}$ converges.

Proof. First of all, we can say that y_n exists for all $n \in \mathbb{N}$ because the **Axiom of Completeness** says that any set that is bounded above has a supremum, and since our sequence (a_n) is bounded, every set $A_n = \{a_k : k \geq n\}$ is also bounded.

Since each set A_n is bounded, the sequence of suprema (y_n) must also be bounded, since the least upper bound must be smaller than or equal to a general bound.

We claim that the sequence (y_n) must be monotone decreasing: Compare y_n to y_{n+1} .

$$\begin{aligned} y_n &= \sup A_n = \sup\{a_n, a_{n+1}, a_{n+2}, a_{n+3}, \dots\} \\ y_{n+1} &= \sup A_{n+1} = \sup\{a_{n+1}, a_{n+2}, a_{n+3}, \dots\} \end{aligned}$$

Since $A_{n+1} \subseteq A_n$, we must have that $\sup A_{n+1} \leq \sup A_n$, since any upper bound for A_n must also necessarily be an upper bound for A_{n+1} . So $y_{n+1} \leq y_n$ for all $n \in \mathbb{N}$.

Thus since (y_n) is monotone and bounded, it converges by the **Monotone Convergence Theorem**. □

4. (5 points) Sketch a proof that $\mathbb{R}$ is uncountable. You may either use Cantor's diagonalization argument or the Nested Interval Property.

For your sketch, you may be informal in your language but please make sure you specify the method used, any construction involved, and why the conclusion follows. Please use actual words, I do not just mean "sketch a picture" (but by all means you may sketch a picture if you wish).

Bonus (5 points): Sketch both proofs.

Cantor's Diagonalization Argument:

Use contradiction. Suppose we've enumerated the real numbers, $x_1, x_2, x_3, \dots$. We aim to construct a number x that's not on the list.

Consider their decimal expansions. Let the first digit after the decimal point of x be different than the first digit after the decimal point of x_1 . Similarly, let the second digit of x be different than the second digit of x_2 . Continue this process.

x can't be on the list: For all $n \in \mathbb{N}$, $x \neq x_n$ because the n th decimal digit would be different in x and x_n .

So $\mathbb{R}$ must be uncountable.

Nested Interval Property:

Use contradiction. Suppose we've enumerated the real numbers, $x_1, x_2, x_3, \dots$. We aim to construct a number that's not on the list.

Construct a sequence of nested closed intervals. Let I_1 be any interval not containing x_1 . Let $I_2 \subseteq I_1$ be any interval not containing x_2 . Continue this process.

The Nested Interval Property tells us that the intersection is nonempty, so there exists $x \in \bigcap_{n=1}^{\infty} I_n$.

But x can't be on the list: for all $n \in \mathbb{N}$, $x \in I_n$. But we constructed I_n so it does not include x_n . So no element on the list is in every interval.

So $\mathbb{R}$ must be uncountable.

5. (10 points) Prove that every Cauchy sequence converges.

Proof. Let (a_n) be a Cauchy sequence.

Since all Cauchy sequences are bounded, we apply the **Bolzano-Weierstrass** theorem, and we have a convergent subsequence $(a_{n_k}) \rightarrow a$.

We want to prove that $(a_n) \rightarrow a$.

Let $\varepsilon > 0$ be arbitrary.

Because (a_n) is Cauchy, there exists an $N \in \mathbb{N}$ such that $\forall m, n \geq N, |a_n - a_m| < \varepsilon/2$.

Because $(a_{n_k}) \rightarrow a$, we can find a specific a_{n_k} such that $n_k \geq N$ and $|a_{n_k} - a| < \varepsilon/2$.

Let $n \geq N$. Therefore

$$|a_n - a| = |a_n - a_{n_k} + a_{n_k} - a| \leq |a_n - a_{n_k}| + |a_{n_k} - a| < \varepsilon/2 + \varepsilon/2 = \varepsilon$$

So by definition, (a_n) converges to a . □

6. (10 points) Describe the iterative process we can perform to create an infinite sequence of nested intervals in order to use the Nested Interval Property to prove the Axiom of Completeness, inspired by the proof of the Bolzano-Weierstrass Theorem. (How do we create the first interval? How do we create each subsequent interval? How will this help us prove the AoC? A full proof is not needed, just an explanation of the technique.)

To prove the Axiom of Completeness, we need to show that any set A that is bounded above has a supremum. So we need to “zoom in” on a supremum using nested intervals.

Create the first interval $I_1 = [a_1, b_1]$ by choosing $a_1 \in A$, and b_1 is an upper bound for A .

Split this interval in half, so we have $[a_1, x]$ and $[x, b_1]$ where $x = \frac{b_1 + a_1}{2}$. We want to pick the interval that must contain the supremum.

- If x is an upper bound for A , choose $I_2 = [a_1, x]$.
- If x is not an upper bound for A , choose $I_2 = [x, b_1]$.

Either way, the supremum (least upper bound) will be in the interval we picked.

Iterate this process infinitely many times.

The Nested Interval Property says that there is something in the infinite intersection, so there is something common in each interval we picked. It turns out that this is the supremum!